# EXPERIMENT – 6

**AIM :**

To write a Python program to simulate the Vacuum Cleaner Problem by cleaning dirty locations in an environment..

**ALGORITHM:**

1. Start.
2. Initialize the environment with clean and dirty locations.
3. Place the vacuum cleaner in the initial location.
4. Check whether the current location is dirty.
5. If dirty, clean the location.
6. If clean, move to the next location.
7. Repeat until all locations are clean.
8. Display the final environment status.
9. Stop.

**PROGRAM :**

*untitled (3.14.3)*

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```
def vacuum_cleaner(room_a, room_b, position):
    rooms = {"A": room_a, "B": room_b}

    while "Dirty" in rooms.values():
        print("Vacuum at", position)

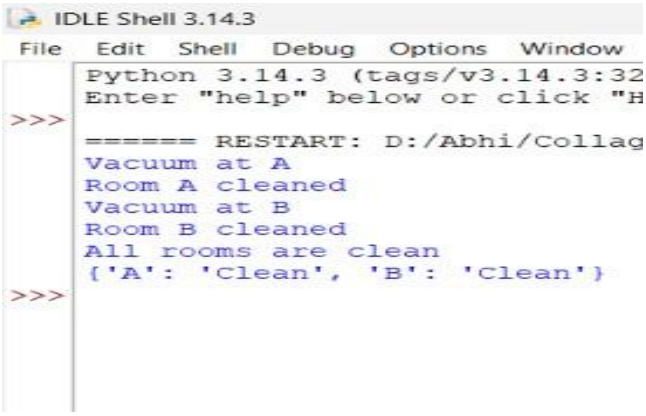
        if rooms[position] == "Dirty":
            rooms[position] = "Clean"
            print("Room", position, "cleaned")

        position = "B" if position == "A" else "A"

    print("All rooms are clean")
    print(rooms)

vacuum_cleaner("Dirty", "Dirty", "A")
```

**OUTPUT:**




**RESULT:**

Thus, the Python program to simulate the Vacuum Cleaner Problem was executed successfully, and all the dirty locations were cleaned.